Technical Data sheet

LoRa Arduino Compatible EVK

Author: Nikunj Mehta

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Preface

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Revision history

Name	Description of Change	Date	Version
Nikunj Mehta	Initial version	29/03/2022	0.1

Approval

Name	Position	Signature	Date
Himanshu Ankleshwaria	Manager	Email approval received (must be stored in document library with Approved version)	
Lalit Shah	Manager	Email approval received (must be stored in document library with Approved version)	

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1. Introduction

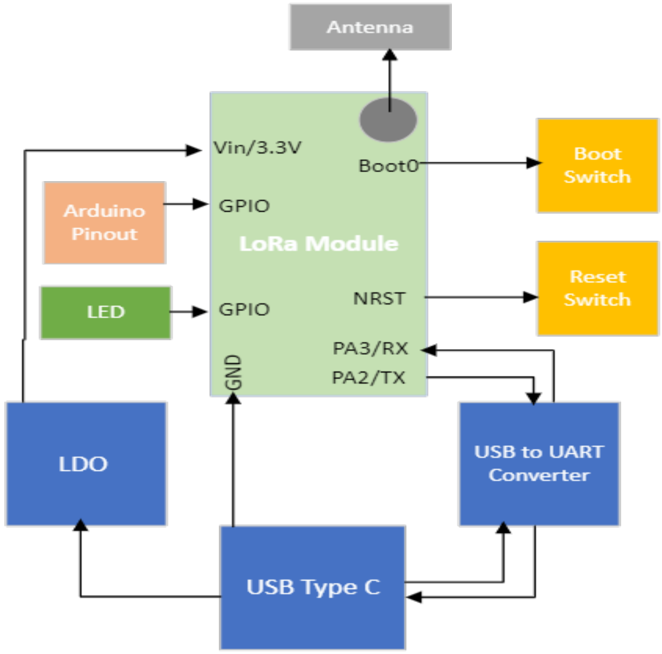
The LoRa Arduino Compatible EVK is based on latest chipset STM32WL SoC which has Embedded LoRaWAN Transceiver for wide range of Smart Applications and high-performance STM32WLE5JC Arm® Cortex®-M4 32-bit RISC core operating at a frequency of up to 48 MHz. This core implements a full set of DSP instructions and an independent memory protection unit (MPU). It is useful when long range is required as it operates at 868 & 915 MHz Band and uses Lo–Ra(Long-Range) Protocol for communication. It offers long range, low power and secure data transmission for M2M and IoT applications. which supports LoRa, (G)MSK and (G)FSK modulations , longer communication distance and stronger anti-interference ability.

LoRa Arduino Compatible EVK provide Arduino nano Pinout compatible option to enhance open source platform on LoRa IOT platform and Arduino programming support on STM32WLE5JC chip set.

1.1. Brief Specifications

LoRa Compliance	LoRa Compliance and refer “NLN500a_Module_Datasheet_V2.0” document for more details.
Power Supply	USB Type C connector option for 5V input
Arduino Compatibility	Support of 30 Pin 2.54mm Pitch header compatible with Arduino Nano pin out.
Software Development Environment support	1) Arduino IDE 2) STM32 Cube IDE
LED indication	LED indication for LoRa communication and Power detection.
Programming	1) Programming over USB (USB to UART on board converter used) 2) SWDIO Programming (ST Programming loader option)
Mode	1) User Mode – By Default running on this mode 2) Programming Mode - For Programming over USB need to press Boot button (boot pin as high) and then press RESET button (for RESET the Device).
Operating Temperature	industrial Temperature range -40 to +85°C (NLN500a having environment Characteristic with temperature range of -20 ºC to 75 ºC)
Battery Operated	Device also have Battery operated support connect 3.7V Battery terminal on 27th pin and GND terminal on 29th pin.

1.2. Block Diagram



2. Key Features

- Compact size 63 X 25.00 mm
- Frequency band Supports EU863-870, US902-928, IN865
- USB Type C connector option for 5V input
- Support of 30 Pin header compatible with Arduino Nano pinout
- Programming over USB port using Arduino IDE (Open source)so, no external programmer required (Compatible with External STM Programmer).
- LED indication for LoRa communication and Power detection.
- Receiver Sensitivity: -122dBm @125K/SF7 , -136dBm @125K/SF12
- UFL Antenna Connector
- Range up to 10 km (Line of Sight)
- Range of several km in urban environment
- Output power level up to 21 dBm

3. Applications

The applications include, but are not limited to:

- Logistic Tracking
- Smart Metering
- Smart Agriculture
- Home, Building, Industrial automation
- Wireless Sensors
- Remote Control
- M2M, IoT

4. Specification

4.1. Functional Characteristics

4.1.1. CPU:

- High-performance STM32WLE5JC Arm® Cortex®-M4 32-bit RISC core operating at a frequency of up to 48 MHz.

4.1.2. Memory:

- Up to 256-Kbyte Flash memory
- Up to 64-Kbyte RAM

4.1.3. Clock sources:

- External HSE Clock 32 MHz TCXO
- External 32 kHz oscillator for RTC with calibration

4.2. RF Characteristics

Parameter	NLN500a
Transmit Power	+21 dBm
Receive Sensitivity (1% PER)	–122 dBm (125 kHz BW, spreading factor 7) and –136 dBm (125 kHz BW, spreading factor 12)
RF Range(Line-Of-Sight)	10KM*
Frequency Band	EU 863-870,US 902-928,IN865
RF Data Rate	0.013 to 17.4 Kbit/s

Note:
*Range figure estimates are based on ideal conditions with limited sources of interference. Actual range will vary based on transmitting power, type of transmitting antenna used, height of transmitting antenna, height of receiving antenna, orientation of transmitter and receiver, weather conditions, interference sources in the area, and terrain between receiver and transmitter.

4.3. Electrical Characteristic

4.3.1. Absolute Max. Rating

Symbol	Description	Min	Max	Unit
VBUS	External main supply voltage	-0.3	+5.2	V

4.3.2. Recommended Operating Conditions

Symbol	Description	Min	Typ	Max	Units
VBUS	Standard operating voltage feed through USB Type C	4.8	5.0	5.2	V

4.3.3. Current Consumption

Mode	Current	Condition
Radio in transmit (HP mode)	110mA	868 MHz,21 dBm,3.3 V,SF7/SF12,125KHz
Radio in receive	10mA	Rx boosted, LoRa 125 kHz, SMPS ON
Standby (+ RTC)	1.6μA	VDD = 3 V
Deep-Sleep mode	1μA	Sleep with cold start (All blocks off)

4.3.4. Environmental Characteristic

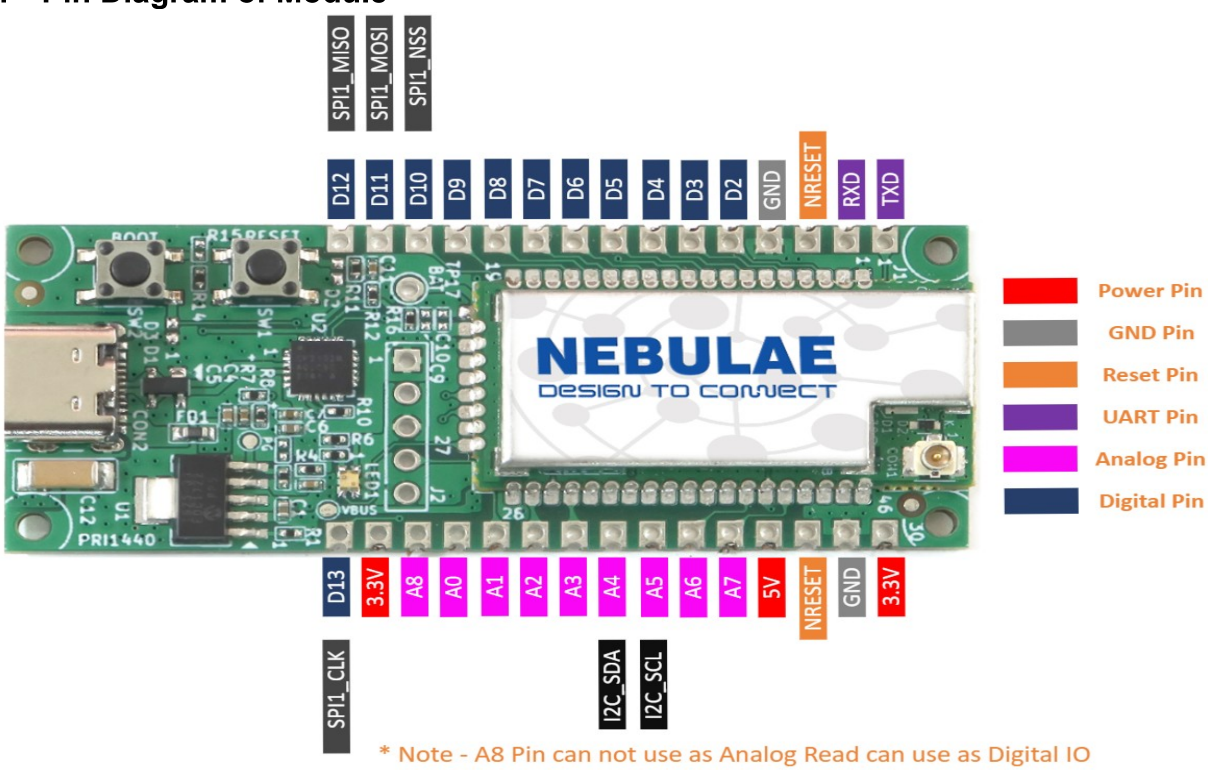
- Operating Temperature: -20 ºC to 75 ºC
- Humidity: 95%

4.3.5. ESD Details of STM32WLE Transceiver Chip

Symbol	Parameter	Conditions	Class	Max.	Units
V_{ESD}	Electrostatic discharge voltage	HBM	2	2000	V
V_{ESD}	Electrostatic discharge voltage	CDM	C2a	500	V

5. Pin Diagram

5.1. Pin Diagram of Module



5.2.Pin Details of the module

Pin #	Pin Name of EVK	Pin Name of MCU	Pin Type	Pin Function/Alternate Functions
1	TXD	PB6	I/O	LPTIM1_ETR, I2C1_SCL, USART1_TX, TIM16_CH1N, CM4_EVENTOUT
2	RXD	PA10	I/O	RTC_REFIN, TIM1_CH3, I2C1_SDA, SPI2_MOSI/I2S2_SD, USART1_RX, DEBUG_RF_HSE32RDY, TIM17_BKIN, CM4_EVENTOUT
3	NRESET	NRST	I/O	Reset
4	GND	GND	S	Ground
5	D2	PB12	I/O	TIM1_BKIN, I2C3_SMBA, SPI2_NSS/I2S2_WS, LPUART1_RTS, CM4_EVENTOUT
6	D3	PA0	I/O	TIM2_CH1, I2C3_SMBA, I2S_CKIN, USART2_CTS, COMP1_OUT, TIM2_ETR, CM4_EVENTOUT
7	D4	PB13	I/O	TIM1_CH1N, I2C3_SCL, SPI2_SCK/I2S2_CK, LPUART1_CTS, CM4_EVENTOUT
8	D5	PA11	I/O	TIM1_CH4, TIM1_BKIN2, LPTIM3_ETR, I2C2_SDA, SPI1_MISO, USART1_CTS, DEBUG_RF_NRESET, CM4_EVENTOUT
9	D6	PC13	I/O	CM4_EVENTOUT
10	D7	PC3	I/O	LPTIM1_ETR, SPI2_MOSI/I2S2_SD, LPTIM2_ETR, CM4_EVENTOUT
11	D8	PC2	I/O	LPTIM1_IN2, SPI2_MISO, CM4_EVENTOUT
12	D9	PB15	I/O	TIM1_CH3N, I2C2_SCL, SPI2_MOSI/I2S2_SD, CM4_EVENTOUT
13	D10	PA4	I/O	RTC_OUT2, LPTIM1_OUT, SPI1_NSS, USART2_CK, DEBUG_SUBGHZSPI_ NSSOUT, LPTIM2_OUT, CM4_EVENTOUT
14	D11	PA12	I/O	TIM1_ETR, LPTIM3_IN1,

				I2C2_SCL, SPI1_MOSI, RF_BUSY, USART1_RTS, CM4_EVENTOUT
15	D12	PA6	I/O	TIM1_BKIN, I2C2_SMBA, SPI1_MISO, LPUART1_CTS, DEBUG_SUBGHZSPI_ MISOOUT, TIM16_CH1, CM4_EVENTOUT
16	D13	PB3	I/O	JTDO/TRACESWO, TIM2_CH2, SPI1_SCK, RF_IRQ0, USART1_RTS, CM4_EVENTOUT
17	3.3V	3V3	S	POWER
18	A8	PC5	I/O	CM4_EVENTOUT
19	A0	PB7	I/O	LPTIM1_IN2, TIM1_BKIN, I2C1_SDA, USART1_RX, TIM17_CH1N, CM4_EVENTOUT
20	A1	PB1	I/O	LPUART1_RTS_DE, LPTIM2_IN1, CM4_EVENTOUT
21	A2	PC1	I/O	LPTIM1_OUT, SPI2_MOSI/I2S2_SD, I2C3_SDA, LPUART1_TX, CM4_EVENTOUT
22	A3	PA9	I/O	TIM1_CH2, SPI2_NSS/I2S2_WS, I2C1_SCL, SPI2_SCK/I2S2_CK, USART1_TX, CM4_EVENTOUT
23	A4	PB9	I/O	TIM1_CH3N, I2C1_SDA, SPI2_NSS/I2S2_WS, IR_OUT, TIM17_CH1, CM4_EVENTOUT
24	A5	PB8	I/O	TIM1_CH2N, I2C1_SCL, RF_IRQ2, TIM16_CH1, CM4_EVENTOUT
25	A6	PA5	I/O	TIM2_CH1, TIM2_ETR, SPI2_MISO, SPI1_SCK, DEBUG_SUBGHZSPI_ SCKOUT, LPTIM2_ETR, CM4_EVENTOUT
26	A7	PA8	I/O	MCO, TIM1_CH1, SPI2_SCK/I2S2_CK, USART1_CK, LPTIM2_OUT, CM4_EVENTOUT
27	5V	-	-	-
28	NRESET	NRST	I/O	Reset
29	GND	GND	S	Ground
30	3.3V	3V3	S	POWER